Material Traceability Report: Ti6Al4V Femoral Stem SLM Production

Document ID: MTR-AM-2024-087 **Report Date:** October 26, 2024 **Process:** Selective Laser Melting (SLM) **Product:** Ti6Al4V Femoral Stem (Medical Device) **Batch:** FS-241015-B20 **Quantity:** 20 units **Facility:** Advanced Orthopedic Manufacturing Center, Building 3

Executive Summary

This report documents the complete material traceability for production batch FS-241015-B20, comprising twenty (20) medical-grade titanium alloy femoral stems manufactured via Selective Laser Melting technology. The material accounting covers the complete production cycle from raw material receipt through final product disposition, verifying mass balance and providing full traceability for quality assurance and regulatory compliance purposes.

All materials used in this production batch conform to established medical device manufacturing standards and are documented with appropriate lot traceability. The material balance demonstrates efficient utilization of input materials with comprehensive tracking of all output streams.

1.0 Material Receipt and Verification

1.1 Raw Material Incoming Inspection

The titanium alloy powder used in this production batch was received under controlled conditions with complete documentation:

Material: Ti6Al4V ELI Grade 23 Powder (Gas Atomized) **Supplier:** Titanium Metals Corporation **Certificate Number:** CofA-89432-AM **Lot Number:** TM-241008-PA **Initial Quantity Received:** 25.00 kg **Quantity Allocated to Batch FS-241015-B20:** 20.83 kg

Material verification testing was performed upon receipt, including: - Chemical composition analysis (Verified per ASTM F3001) - Particle size distribution (D10: 25.2μm, D50: 37.8μm, D90: 52.1μm) - Flow rate: 28.3 s/50g - Apparent density: 2.45 g/cm³ - Moisture content: <0.02%

All test results confirmed material compliance with established specifications for medical device additive manufacturing.

1.2 Process Gas Supply

High-purity argon gas was supplied for both chamber preparation and the building phase:

Gas Supplier: Linde Healthcare Gases **Argon Purity:** 99.999% (Medical Grade) **Certificate:** MG-Ar-89475 **Total Gas Consumption:** 28.97 kg

2.0 Build Process Documentation

2.1 Machine Preparation and Parameters

Manufacturing Equipment: EOS M 290-400W **Machine ID:** AM-07-B3 **Build Job ID:** BJ-241015-087 **Total Build Time:** 61.35 hours **Laser Power Setting:** 400 W **Build Platform Temperature:** 80°C **Chamber Atmosphere:** Argon (<100 ppm O_2)

The build chamber was prepared following standard operating procedure SOP-AM-003, which includes: - Initial chamber evacuation to 0.1 mbar - Argon flooding to achieve inert atmosphere - Continuous argon flow maintenance during building phase

2.2 Material Consumption Summary

The following table summarizes all material inputs consumed during the production process:

Material Type	Specification	Quantity	Purpose	Lot/Traceability
Ti6Al4V Powder	GA, 15-53μm	20.83 kg	Primary build material	TM-241008-PA
Argon Gas	Medical Grade, 99.999%	3.03 kg	Chamber flooding	MG-Ar-89475
Argon Gas	Medical Grade, 99.999%	25.94 kg	Building phase	MG-Ar-89475
Electricity	400V/3-phase	147.26 kWh	Machine operation	Utility Meter AM-07

Energy Context: The total electricity consumption of 147.26 kWh was recorded by the dedicated utility meter for machine AM-07. This aligns with the recorded build time of 61.35 hours at an average machine power consumption of 2.4 kW.

3.0 Output Material Disposition

3.1 Product Output

The primary output of the manufacturing process consists of twenty (20) femoral stem components including integrated support structures:

Product: Ti6Al4V Femoral Stem with Support Structures **Total Product Mass:** 1.77 kg **Unit Count:** 20 stems **Average Mass per Stem (with supports):** 88.5 g

All components remain attached to the build platform at this stage, with support structures intact. Each component is identifiable by its position on the build platform as documented in build file BJ-241015-087.

3.2 Material Recovery and Recycling

The unmelted powder recovered from the build chamber is processed for potential reuse:

Material Stream	Quantity	Disposition	Quality Status
Unmelted Powder	18.99 kg	Sent to Powder Recycling	Recyclable
Support Structures & Minor Losses	0.019 kg	Sent to Metal Recycling	Scrap

The recovered powder undergoes sieving and characterization before being evaluated for reuse in subsequent production batches. This material is currently held in quarantine pending quality assessment.

3.3 Waste Streams

Non-recoverable waste materials are documented and disposed of according to established procedures:

Waste Stream	Quantity	Disposition Method	Documentation
Filter-Captured Powder	0.0208 kg	Landfill Disposal	Waste Manifest WM-241029-15

The filter-captured powder represents fine particulate matter collected by the machine filtration system during the build process and is not suitable for recycling due to potential contamination and altered material properties.

4.0 Material Balance Verification

4.1 Mass Balance Calculation

The complete material accounting for batch FS-241015-B20 demonstrates the following mass balance:

Total Input Mass: - Ti6Al4V Powder: 20.83 kg - Process Gases: 28.97 kg (Note: Gases are process media, not incorporated into solid materials)

Total Output Mass (Solid Materials): - Product Components: 1.77 kg - Recovered Powder: 18.99 kg - Support Structures & Losses: 0.019 kg - Filter Waste: 0.0208 kg - **Total Solid Output:** 20.7998 kg

Mass Balance Reconciliation: - Input Powder: 20.8300 kg - Output Solids: 20.7998 kg - **Variance:** 0.0302 kg (0.15%)

The minor mass variance of 0.15% falls within the acceptable tolerance range of ±0.2% established for this manufacturing process, attributed to measurement uncertainty and minor material handling losses.

4.2 Material Yield Analysis

Powder Utilization Efficiency: - Powder converted to product: 1.77 kg (8.5% of input powder) - Powder recovered for potential reuse: 18.99 kg (91.1% of input powder) - Powder lost to waste streams: 0.0398 kg (0.19% of input powder) - Unaccounted variance: 0.0302 kg (0.15% of input powder)

Historical Comparison: Previous batch FS-240930-B20 showed similar yield characteristics with 8.6% conversion to product and 91.0% powder recovery.

5.0 Traceability Records

5.1 Component-Level Traceability

Each femoral stem produced in this batch maintains individual traceability through:

- Build platform position coordinates
- Unique serialization upon removal from build platform
- Individual mass records post-support removal
- Non-destructive testing results per component

5.2 Documentation Chain

The complete material traceability is supported by the following documentation:

- Raw Material Certificate: CofA-89432-AM
- Gas Quality Certificate: MG-Ar-89475
- Build Process Record: BPR-241015-087
- Material Weighing Records: MWR-241029-01 through MWR-241029-08
- Waste Disposal Documentation: WM-241029-15
- Quality Inspection Records: QIR-241029-22

6.0 Quality Assurance Verification

All material handling and processing activities were performed in compliance with established quality procedures:

- **SOP-AM-003:** Powder Handling and Storage
- **SOP-AM-007:** Build Chamber Preparation
- **SOP-AM-012:** Powder Recovery and Recycling
- **SOP-AM-015:** Waste Material Disposal

The material traceability documented in this report confirms compliance with regulatory requirements for medical device manufacturing, including FDA Quality System Regulations and ISO 13485 standards.

7.0 Conclusion

The material traceability for production batch FS-241015-B20 has been fully documented and verified. All material inputs have been accounted for with appropriate disposition of outputs. The mass balance demonstrates efficient material utilization with minimal losses. The complete traceability chain from raw material through final components has been established and documented for regulatory compliance and quality assurance purposes.

This batch has been approved to proceed to the next manufacturing stage (support structure removal and initial inspection) based on satisfactory material traceability documentation.

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